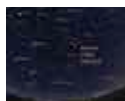




James Webb Telescope

The James Webb Space Telescope successfully lifts off on a historic mission. <https://digit.in/jan22-39>



Asteroid passing Earth

An asteroid 2.5 times the height of the Empire State Building will pass Earth on January 18th. <https://digit.in/jan22-40>

our atmosphere but also helps speed up the transition to alternative fuels.

THE MUSK TWEET

The announcement was made via Musk's official Twitter account, something he is notoriously known for, where he tweeted about SpaceX starting a program to take CO₂ out of the atmosphere and turn it into rocket fuel.



He also announced that this will be important for the Mars space missions alluding to the carbon capture program's intent to manufacture propellants on Mars as the red planet's atmosphere is predominantly composed of carbon dioxide.

The idea is to refuel the spacecraft beyond the confines of Earth making it a crucial part of Elon Musk's long-term vision for humans to become an interplanetary species. This also opens up the possibility of fuel depots on Mars itself, making return trips a distinct and achievable possibility. Combine that with the reusable launch vehicles SpaceX have been working on and you can see how Elon's long-term plans for colonising Mars are fitting into place.

THE NEED TO COLONISE MARS AND BEYOND

When it involves colonising space, the argument has always been

that becoming an interplanetary species will guarantee humanity's survival. In addition to Musk, the argument has been made by such luminaries as Stephen Hawking, J. Richard Gott and others. But, that needs to be done in a more sustainable way and alternative fuels might be the way to go.

OTHER CARBON CAPTURE METHODS

World political leaders are attempting to neutralise carbon emissions by carbon capture methods like planting more trees which has been the traditional and go-to method since years now. The problem with that is that it isn't as efficient or as quick as we need it to be. However, new technologies are being developed to remove carbon at a major scale and function as carbon capture systems.

The Orca Plant in Iceland for example, uses direct air capture technology and is expected to remove 4,000 tonnes of carbon from the air each year. Filters in massive fans at the plant capture carbon from the atmosphere and store it below ground in basalt rock where it can be mineralized.

Scientists are also studying the use of microbes which turns carbon dioxide into fuel. They have engineered a kind of bacterium that takes the carbon dioxide from the air and turns it into fuel in just one enzymatic step. The bacterium *Rhodospseudomonas palustris* uses sunlight to produce methane and hydrogen inside the bacterium itself. The research seems to show a lot of promise if it is scalable.



Image Source: Universe Today

There is also the Canadian company Carbon Engineering, that's making liquid fuel with carbon dioxide directly removed from the atmosphere and turns it into pipeline-ready compressed CO₂ gas with just the use of energy and water. It can then be combined with non-fossil fuel-generated hydrogen to produce hydrocarbon fuels.

The New York startup, Air Company, has the business of converting Earth's CO₂ into usable products such as perfumes and impurity-free alcohol.

Along with that, NASA has also developed a method of turning CO₂ into fuel using metal oxide thin film devices that convert gas into fuel. The device can be used in industrial processes, converting carbon dioxide into fuel such as methane.

Carbon capture is something important to Musk. His foundation funded the XPRIZE Carbon Removal competition with \$100 million, which aimed to fight climate change and rebalance our planet's carbon cycle. So far, only \$5 million of that funding has been used.

Musk, along with his SpaceX team, are hopeful about finding an innovative way to convert atmospheric carbon dioxide into rocket fuel. If this goal is achieved it will open the doors for other greenhouse gases being taken from the air and recycled for other purposes.

There's additionally the prospect of leveraging the abundant resources of space to meet the needs of Earth while at the same time reducing the environmental impact we have had on the world. In this regard, big corporations and businesses (which happen to be majorly responsible for the dire situation we are in) need to pull their socks up by adopting and scaling up these innovative carbon capture methods to offset the damage that is already done. 